

Yeshua Macwan

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EDUCATION

California State University, Sacramento
B.S. in Electrical and Electronics Engineering
Minor: Physics and Computer Science

Sacramento, CA
Expected Dec, 2028

SKILLS

Languages: C/C++, Java, MATLAB, Verilog
Tools: NI Multisim, VS Code, Eclipse, Arduino IDE, FPGA tools, HFSS
Hardware: Breadboarding, soldering, multimeter, oscilloscope, circuit debugging, vector network analyzer

PROJECT EXPERIENCE

Hardware Based Dolphin Attack Detection System
Team Engineering Project

Sacramento, CA
Apr 2026 – May 2026

- Designed an ultrasonic attack detection circuit to identify hidden voice-command attacks.
- Implemented RL filtering and two-stage amplification for ultrasonic signal processing.
- Built NAND-gate and transistor circuitry to disable microphone input during attacks.
- Analyzed oscilloscope measurements to evaluate signal attenuation and circuit behavior.

Slot-Fed Planar Antenna Design
Team Electromagnetics Project

Sacramento, CA
Apr 2026 – May 2026

- Designed and simulated a slot-fed planar antenna in HFSS for RF communication applications.
- Calculated antenna dimensions and impedance matching to support efficient power transfer.
- Fabricated and tested the antenna using VNA measurements to evaluate resonant frequency and bandwidth.
- Compared simulated and measured results to identify fabrication and design limitations.

Logic Design
Digital Systems Project

Sacramento, CA
Sep 2025 – Dec 2025

- Designed and simulated combinational logic circuits including multiplexers, adders, and comparators.
- Wrote Verilog modules and verified outputs using truth tables and simulation waveforms.
- Implemented and debugged circuits on breadboards using systematic probing and signal verification.
- Compared theoretical predictions, simulation results, and measured outputs to validate circuit behavior.

Data Structures & Algorithms
Programming Project

Sacramento, CA
Nov 2026 – Dec 2026

- Built a Java console app using a heap-based PriorityQueue.
- Implemented add, peek, print, and delete queue operations.
- Enabled dynamic sorting with custom Comparator logic.
- Added input validation for dates and recommendation scores.

CAD Parts Modeling
CAD Modeling (Onshape)

Sacramento, CA

- Built 3D models in Onshape from engineering drawings using precise dimensions and design intent.
- Applied features such as revolve, mirror, chamfer, and pattern to create accurate mechanical parts.
- Developed foundational CAD modeling skills through dimensioned part creation and feature-based design

3D Printing
3D Printing & Slicer Preparation

Sacramento, CA

- Imported online models into Ultimaker Cura and prepared them for printing
- Separated parts, adjusted orientation, and selected support settings based on part geometry.
- Used tree supports when appropriate to reduce PLA usage and simplify support removal
- Troubleshoot print issues including layer shifting to improve print quality and consistency

WORK EXPERIENCE

Boys and Girls Club
Digital Navigator Intern

Sacramento, CA
Sep 2024 – Apr 2025

- Taught K–6 students about online safety, responsible browsing, and internet use.
- Assisted in creating content for digital literacy workshops and activities.
- Supported student learning by completing digital tasks and communicating clearly with students and staff.

COURSE EXPERIENCE

Circuit Analysis

Analytical Engineering Coursework

Sacramento, CA

Sep 2025 – Dec 2025

- Applied nodal, mesh, Thevenin, and Norton analysis to electrical networks.
- Analyzed first- and second-order circuits, including transient responses.
- Performed hand calculations to evaluate and verify circuit behavior.
- Solved problems using multiple systematic solution approaches.

Electricity & Magnetism

Academic Laboratory Experience

Sacramento, CA

Jan 2025 – May 2025

- Measured voltage and current using multimeters and oscilloscopes.
- Built and tested RC, RL, RLC, and LC circuits and recorded responses.
- Interpreted schematics and assembled circuits on breadboards.
- Verified lab results using hand calculations and data comparisons.